

# RAKESH KUMAR

Full-Stack Developer | MERN Stack Engineer

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## PROFESSIONAL SUMMARY

Full-stack developer specializing in systems with real engineering depth - built an incident root-cause engine benchmarked at 89.5% accuracy versus a 73.7% baseline, a constraint-satisfaction scheduler using MRV/LCV backtracking, and real-time collaborative tools with CRDT-based sync.

## EDUCATION

### Bachelor of Technology (B.Tech), Computer Science Engineering

May 2026

Parul University, Vadodara

**Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), TypeScript, Python, C++, SQL

**Frontend:** React.js, Next.js, Redux, React Router, Tailwind CSS, HTML5, CSS3, D3.js

**Backend & APIs:** Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT, RBAC

**Databases:** MongoDB, Mongoose, PostgreSQL, MySQL, Redis, pgvector

**AI / LLM:** Gemini API, Vector Embeddings, Prompt Engineering, Deterministic Evaluation

**Real-Time & Data:** D3.js, Yjs (CRDT), WebSocket, WebAssembly

**Tools & Deployment:** Git, GitHub, Postman, Vercel, Render, npm

## PROJECTS

### TRACE - Temporal Root-Cause Analysis & Causal Engine | [GitHub](#) | [Live](#)

Python | FastAPI | PostgreSQL (pgvector) | Gemini API | Next.js | TypeScript | D3.js

- Built an incident investigation engine that correlates logs, metrics, traces, and deployment events to identify root causes in synthetic production scenarios.
- Designed a deterministic hypothesis-scoring system using temporal, causal, dependency, and evidence signals, paired with an LLM critique loop that attempts to falsify the leading diagnosis.
- Achieved 89.5% root-cause accuracy across 19 test incidents with hidden root causes, outperforming a 73.7% single-prompt LLM baseline; added a deterministic trend comparison check after identifying an adversarial failure case.

### CHRONOS - Constraint-Based Timetable Scheduling Engine | [GitHub](#) | [Live](#)

React | Node.js | PostgreSQL | Gemini API | D3.js | GSAP

- Built a timetable scheduling engine that solves constraint satisfaction problems using backtracking with MRV and LCV heuristics.
- Developed an interactive D3.js visualization comparing naive and heuristic-guided searches on identical scheduling inputs, showing explored states, backtracking, and pruning in real time.
- Integrated the Gemini API to translate natural-language scheduling requirements into structured constraint inputs.

### SyncPad - Real-Time Collaborative Code Editor | [GitHub](#) | [Live](#)

React | TypeScript | Node.js | Yjs (CRDT) | Monaco Editor | WebSocket | Pyodide (WASM)

- Built a real-time collaborative editor with conflict-free synchronization using Yjs CRDTs, including live multi-cursor presence and awareness.
- Implemented sandboxed browser-based execution for JavaScript and TypeScript using timeout-protected Web Workers and for Python using Pyodide (WebAssembly), eliminating reliance on third-party execution APIs.
- Deployed a Node.js synchronization server using y-websocket on Render and a React/Vite client on Vercel.

## PROFESSIONAL EXPERIENCE

### Software Development Intern | Codetech IT Solutions | Remote

Jan 2026 - Apr 2026

- Replaced manual spreadsheet tracking with a role-aware employee management system using Node.js, Express.js, and MongoDB for employee, manager, and admin roles.
- Secured API access across all 3 user tiers with JWT authentication and RBAC middleware, preventing unauthorized cross-role data access.
- Built and tested CRUD REST APIs for 3 core modules - employee, department, and role - with validation and RBAC, forming the system's primary data access layer.

## CERTIFICATIONS & ACTIVITIES

- SQL (Advanced) - HackerRank | C++ Programming - OpenEDG/Cisco | Introduction to IoT - NPTEL (IIT Kharagpur), Elite
- AMENTIS: Chartering the Unknown Hackathon - IEEE GTBIT Student Branch | CodeKshetra Coding Contest - GeeksforGeeks (GD Goenka)